

Rahul Tammalla

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Professional Summary

Data Scientist specializing in statistical modeling, predictive analytics, and end-to-end ML pipelines, from EDA and feature engineering through model validation and production-oriented delivery. M.S. Applied Data Science (GPA: 3.8) with research-grade grounding in hypothesis testing, classification, and NLP across clinical, public-safety, and enterprise analytics domains.

Education

Clarkson University <i>M.S., Applied Data Science, Merit Scholar, GPA: 3.8/4.0 (Graduate TA, Probability & Statistics)</i>	Potsdam, NY Aug 2024 – May 2026
Amrita Vishwa Vidyapeetham <i>Bachelor's in Computer Science and Engineering</i>	Kerala, IND May 2017 – Jun 2021

Skills

Core ML & Statistics: Python, SQL, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Random Forest, SVM, Statistical Modeling, Hyperparameter Tuning, Multivariate Imputation
Deep Learning & Retrieval: PyTorch, FAISS, Embeddings, Feature Engineering, HNSW, IVF
Data Analysis & BI: EDA, Tableau, Power BI, ROC-AUC, k-fold CV, MLflow, Precision-Recall
Cloud & Data Infra: AWS, Snowflake, Databricks, PySpark, Airflow, CI/CD

Work Experience

BalanX Bio <i>Data Science Intern</i>	East Lansing, MI Jul 2026 – Present
- Designed a sliding-window pattern-detection model for multi-modal emotional signals, clustering observations by valence-arousal dominance distance and cosine similarity to cut one-off false-positive writes 40%. - Built and evaluated a composite retrieval-ranking method combining semantic similarity, emotional alignment, and time decay, comparing 5 approaches across 4 design iterations to lift relevance 20%.	

Clarkson University <i>Statistical Research Assistant</i>	Potsdam, NY Jan 2025 – Jun 2026
- Conducted EDA and statistical analysis across 20M+ NEMSIS emergency records using Python, SQL, and hypothesis testing, engineering 70+ features and identifying 15 significant predictors of stroke-related dispatch delays. - Built Python/PySpark pipelines resolving 1M+ missing values through multivariate imputation, then trained and tuned Random Forest, XGBoost, LightGBM, and SVM models using k-fold validation and ROC-AUC evaluation. - Published model results through PostgreSQL and Tableau dashboards, enabling research stakeholders to investigate delay factors, model performance, and high-risk conditions.	

I4DI (Institute for Development Impact) <i>Data Science Intern</i>	Washington, DC May 2025 – Aug 2025
- Optimized GTE embedding inference in PyTorch using dynamic batching and asynchronous data loading, cutting query latency from 5.1s to 2.9s while improving Precision@K 30%. - Implemented FAISS HNSW and IVF indexes with cosine similarity and metadata filtering, evaluating retrieval experiments on Recall@K, MRR, and nDCG with MLflow tracking.	

Cognizant Technology Solutions <i>Data Engineer</i>	Remote Oct 2021 – Aug 2024
- Cleaned and transformed enterprise data on AWS using Databricks and PySpark (schema validation, deduplication, joins), loading 5M+ daily records into Snowflake for SQL-based business-rule validation and aggregation feeding downstream models. - Monitored Airflow pipelines and traced failures through Snowflake and Databricks to source data, adding validation checks and anomaly alerts that cut failures 70% and lifted query performance 30%, surfaced through Tableau/Power BI monitoring dashboards.	

Projects

Customer Revenue & Retention Analytics	Github
- Modeled cohort retention, RFM segmentation, and CLV across 125K orders and 406K funnel events. - Ran hypothesis testing and correlation analysis, fitting a regression forecast on a time-based holdout, achieving an R² of 0.928.	
Hybrid Retrieval & Ranking System	Github
- Built BM25 plus dense FAISS retrieval with cross-encoder reranking, tracked across graded query sets in MLflow. - Evaluated with Precision@K, MRR, and nDCG, reaching 0.700 MRR and retaining the lexical baseline as champion.	
Financial Analysis – SaaS FP&A Control Tower	Github
Built a 48-month budgeting, forecasting, and budget-to-actual model tracking \$5.16M ARR against a \$3.57M YTD loss, modeling Monte Carlo liquidity risk at 59.7% and 5.7x LTV/CAC unit economics.	

Certifications

Certificate of Completion: Claude 101, Anthropic | Jun 2026 • AI Fluency Framework & Foundations, Anthropic | Jun 2026